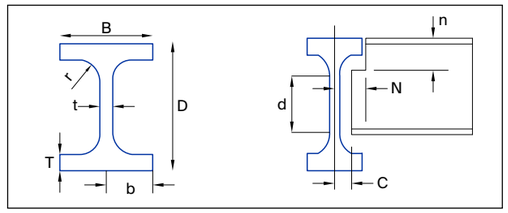
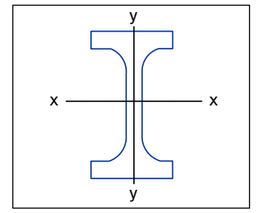
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W21-21x12 1/4 (533x312)	201	300	585.0	319.0	41.4	23.1	12.7	476.8	382	3.85	20.6	14	158	54
	223	331	593.0	322.0	45.5	25.4	12.7	476.6	422	3.54	18.8	15	158	58
	248	370	603.0	324.0	50.5	26.2	12.7	476.6	460	3.21	18.2	15	159	63
	275	409	613.0	327.0	55.6	31.0	12.7	476.4	521	2.94	15.4	18	158	68
W24-24x7 (610x178)	55	81.9	598.7	177.9	12.8	10.0	12.7	547.6	105	6.93	54.6	7	94	26
	61	91.0	602.5	178.4	15.0	10.6	12.7	547.1	116	5.95	51.6	7	94	28
	62	92.3	603.0	178.8	15.0	10.9	12.7	547.6	118	5.96	50.1	7	94	28
W24-24x9 (610x229)	68	101.2	602.6	227.6	14.8	10.5	12.7	547.6	129	7.69	52.2	7	119	28
	76	113	607.6	228.2	17.3	11.1	12.7	547.6	144	6.60	49.3	8	119	30
	84	125.1	612.2	229.0	19.3	11.9	12.7	548.2	158	5.93	46.1	8	119	32
	94	139.9	617.2	230.2	22.1	13.1	12.7	547.6	178	5.21	41.8	9	119	35
	103	153	623.1	228.6	24.9	14.0	12.7	547.9	195	4.59	39.1	9	117	38
	114	171	628.9	229.9	27.9	15.5	12.7	547.7	218	4.12	35.3	10	117	41
	128	191	635.0	231.5	30.9	17.0	12.7	547.8	242	3.75	32.2	11	117	44
	146	217	643.1	234.1	35.1	19.6	12.7	547.5	278	3.33	27.9	12	117	48
	163	243	651.0	235.9	39.1	21.6	12.7	547.4	310	3.02	25.3	13	117	52
	181	270	658.6	238.5	42.9	23.9	12.7	547.4	343	2.78	22.9	14	117	56
	198	295	667.0	240.5	47.0	25.9	12.7	547.6	376	2.56	21.1	15	117	60
	218	325	675.1	243.1	51.0	28.4	12.7	547.7	412	2.38	19.3	16	117	64
	239	356	685.0	245.6	55.9	31.0	12.7	547.8	454	2.20	17.7	18	117	69
W24-24x12 (610x305)	100	149.1	612.4	304.8	19.7	11.8	16.5	540.0	190	7.74	45.8	8	157	36
	120	179	620.2	307.1	23.6	14.1	16.5	540.0	228	6.51	38.3	9	157	40
	160	238.1	635.8	311.4	31.4	18.4	16.5	540.0	303	4.96	29.3	11	157	48
W24-24x12 3/4 (610x324)	104	154.8	611.1	323.8	19.1	12.7	12.7	547.6	198	8.50	43.1	8	166	32
	117	174.1	616.2	325.1	21.6	14.0	12.7	547.6	222	7.53	39.2	9	166	34
	131	194.9	621.8	326.5	24.4	15.4	12.7	547.6	249	6.70	35.6	10	166	37
	146	217.3	628.4	327.7	27.7	16.5	12.7	547.6	277	5.92	33.2	10	166	40
	162	241.1	635.0	329.1	31.0	17.9	12.7	547.6	308	5.31	30.6	11	166	44
	176	261.9	641.1	327.4	34.0	19.1	12.7	547.6	333	4.81	28.7	12	164	47
	192	285.7	646.9	328.9	37.1	20.6	12.7	547.3	363	4.44	26.6	12	164	50
	207	308	653.0	330.5	39.9	22.1	12.7	547.8	392	4.14	24.8	13	164	53
	229	341	661.0	333.0	43.9	24.4	16.5	540.2	435	3.79	22.1	14	164	60
	250	372	669.0	335.0	48.0	26.4	16.5	540.0	475	3.49	20.5	15	164	65
	279	415	679.0	338.0	53.1	29.5	16.5	539.8	530	3.18	18.3	17	164	70
	306	455	689.0	340.0	57.9	32.0	16.5	540.2	579	2.94	16.9	18	164	74

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.38	220700	22450	24.1	7.67	7547	1408	8670	2175	0.890	14.6	16.6	1700
2.40	247700	25390	24.2	7.76	8355	1577	9656	2442	0.890	13.4	19.0	2263
2.43	278900	28710	24.6	7.90	9250	1772	10730	2739	0.894	12.3	21.9	2996
2.45	316900	32530	24.7	7.91	10340	1990	12120	3096	0.890	11.2	25.3	4131
1.87	56020	1210	23.2	3.40	1871	136	2200	219	0.843	48.3	1.04	49.2
1.88	63880	1426	23.5	3.51	2120	160	2480	256	0.850	43.2	1.23	68.6
1.88	64580	1436	23.4	3.49	2142	161	2512	258	0.848	42.8	1.24	71.0
2.07	75780	2915	24.2	4.76	2515	256	2881	400	0.864	43.1	2.52	77.0
2.08	87320	3434	24.6	4.88	2874	301	3281	469	0.870	38.0	2.99	111
2.09	97540	3872	24.8	4.95	3187	338	3639	528	0.872	34.5	3.40	149
2.11	111800	4505	25.0	5.03	3622	391	4142	611	0.875	30.6	3.99	216
2.11	125000	4972	25.3	5.04	4012	435	4595	680	0.878	27.6	4.45	295
2.12	141400	5670	25.4	5.09	4495	493	5167	773	0.878	24.8	5.12	411
2.14	158400	6415	25.6	5.15	4990	554	5757	871	0.879	22.6	5.85	554
2.16	183900	7544	25.7	5.21	5718	644	6643	1019	0.877	20.0	6.97	819
2.18	207900	8606	25.9	5.27	6386	730	7455	1157	0.878	18.2	8.06	1123
2.20	232800	9768	26.1	5.34	7069	819	8299	1304	0.877	16.7	9.26	1495
2.22	259400	10980	26.3	5.41	7778	913	9173	1458	0.878	15.4	10.6	1952
2.24	287700	12330	26.4	5.47	8522	1014	10110	1625	0.877	14.3	12.0	2517
2.27	322200	13950	26.6	5.55	9406	1136	11220	1826	0.877	13.2	13.8	3309
2.39	125900	9308	25.7	7.00	4111	611	4594	937	0.886	32.7	8.17	200
2.41	153000	11410	25.9	7.07	4935	743	5547	1144	0.886	27.7	10.2	340
2.45	209500	15840	26.3	7.23	6589	1017	7486	1574	0.886	21.3	14.5	785
2.47	129200	10790	25.6	7.39	4227	666	4734	1023	0.880	33.6	9.46	196
2.48	147200	12380	25.8	7.47	4776	762	5360	1170	0.882	30.0	10.9	280
2.50	167300	14160	25.9	7.55	5383	867	6057	1335	0.883	26.8	12.6	395
2.51	190800	16260	26.2	7.66	6074	993	6845	1527	0.886	23.9	14.7	559
2.53	215400	18440	26.4	7.74	6784	1121	7670	1726	0.888	21.6	16.8	770
2.53	236600	19950	26.6	7.73	7380	1218	8369	1878	0.890	19.9	18.4	995
2.55	260400	22030	26.8	7.79	8050	1340	9163	2068	0.890	18.4	20.5	1280
2.56	283900	24050	26.9	7.84	8695	1455	9936	2250	0.890	17.2	22.6	1593
2.58	319000	27090	27.1	7.90	9652	1627	11090	2523	0.890	15.7	25.8	2162
2.60	353900	30170	27.3	7.97	10580	1801	12220	2797	0.890	14.5	29.1	2809
2.62	400500	34310	27.5	8.04	11800	2030	13720	3162	0.889	13.2	33.6	3824
2.65	445200	38100	27.7	8.11	12920	2241	15120	3498	0.889	12.2	37.9	4940